

Predictive supply chain for Primary Health Centres.

A pharmacist *speaks* a stock update. Vertex AI forecasts the next 7 days and flags the stock-out. The engine moves the surplus — before a patient is turned away.

THE PROBLEM

Indian public facilities hold **17–51%** of essential medicines. The WHO benchmark is **80%**.

- Stock-outs run **4 to 14 weeks** at a time
- Stock ledgers are **paper** — district consolidation takes weeks
- Districts learn about a shortage **after** patients are turned away

We are not short on procurement money. We are short on **warning**.

Cited: availability range and stock-out duration — “Factors Affecting the Availability and Utilization of Essential Medicines in India: A Systematic Review” (PMC11174260). 80% benchmark per WHO.

They record the past. They never predict the future.



Every e-governance dashboard today is a *rear-view mirror*.
We built a **windshield**.

Speak. Predict. Redistribute.

- A PHC worker **speaks** “பாராசிட்டமால் ஐம்பது” — the ledger updates itself
- Vertex AI forecasts **7-day** demand → a **0–100 stock-out Risk Score** per medicine
- One tap moves surplus from an **overstocked** PHC to a **starved** one
- Supervisors alerted at risk > 80 — FCM push, WhatsApp in production

Three Google AI systems, one feature contract.

SYSTEM	ROLE	ENDPOINT
Vertex AI AutoML	7-day demand forecasting	aiplatform.googleapis.com
Gemini 2.0 Flash	Multilingual voice → structured entry	generativelanguage.googleapis.com
Cloud Speech-to-Text	Tamil · Hindi · English capture	speech_to_text
On-device fallback	Same contract when Vertex is unreachable	no network required

The forecast, the voice parser and the offline fallback all consume the **same engineered features** — so quality never silently changes with connectivity.

End to end, in 90 seconds.

- Voice in Tamil → Firebase write
- Risk card **spikes to 97%**
- Transfer recommendation appears
- One tap → stock moves, alert logged

LIVE CAPTURE

FIREBASE RTDB

OFFLINE FALLBACK

ArogyaChain AI

PHC-001 Thiruvallur Main

DEMO MODE

Command

Facility

Transfers

Evidence

PHC-001

Thiruvallur Main - 8 medicine pairs tracked

2 CRITICAL

Voice capture

Speak the dispensing update – no typing, no register

Tap the mic, then say for example "Paracetamol 50"

தமிழ் தெளிவு English

MANUAL FALLBACK

Paracetamol 50

Used when connectivity drops or speech is unavailable.

Medicines

Sorted by stock-out risk - 7-day forecast

High first

Search medicines

MEDICINE	RISK	DAYS LEFT	STOCK - 7D FC
Amoxicillin 500mg on-device ensemble	CRITICAL 96%	0.6 d Oct 1	10 · 120 reorder 150
Paracetamol 500mg on-device ensemble	CRITICAL 91%	1.7 d Oct 2	40 · 168 reorder 200
Salbutamol 100mcg on-device ensemble	WATCH 41%	9.4 d Oct 10	85 · 63 reorder 110
Iron Folic Syrup on-device ensemble	HEALTHY 27%	37.3 d	410 · 77 reorder 130
Metformin 500mg on-device ensemble	HEALTHY 22%	45.0 d	450 · 70 reorder 120
Amlodipine 5mg on-device ensemble	HEALTHY 18%	40.0 d	320 · 60 reorder 90
Cetirizine 10mg on-device ensemble	HEALTHY 11%	43.3 d	260 · 42 reorder 80
ORS Low Osmolarity	HEALTHY 8%	50.0 d	600 · 95

Beats the rule-based baseline by **16.1%** on a 28-day holdout.

20.93%

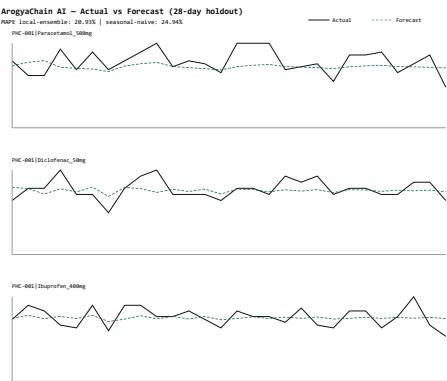
our ensemble forecaster — MAPE

24.94%

seasonal-naïve baseline — MAPE

6,355

forecasts evaluated, 240 series



Measured by `backtest.py` → `data/validation_metrics.json`. Reproducible by a judge in one command.

Stock-outs fall 4,863 → 38. And it survives stress.

99.2%

fully-connected district — upper bound

71.0%

nearest-neighbour only — the honest floor

65,229 → 122

units of unmet demand

DONOR REACH	STOCK-OUTS	REDUCTION	UNMET-DEMAND REDUCTION
1 PHCs	1,408	71.0%	71.2%
2 PHCs	46	99.1%	99.8%
3 PHCs	46	99.1%	99.8%
all PHCs in district	38	99.2%	99.8%

Measured by `impact_sim.py` → `data/impact_metrics.json`. Paired counterfactual: both arms consume the identical RNG stream, so both face identical latent demand. Simulated, not field-measured — we had no pilot, so we measured the mechanism.

Three roles. One system.

PHC pharmacist

Speaks the update instead of maintaining a register. Zero learning curve, works in Tamil.

District Health Officer

Sees which of their PHCs break first — and which can lend before it happens.

State procurement

Sees demand 7 days out, before raising a purchase order at a premium.

The District Health Officer is the buyer.

Voice input is what makes the pharmacist's participation non-negotiable — if it is not zero-effort, the data stops arriving.

Pilot-ready in two weeks, on infrastructure a district already has.

CONCERN	HOW IT IS ANSWERED
Cost	Firebase Spark tier — free for a district pilot. No new hardware.
Connectivity	Offline-first writes plus an on-device forecaster. Degrades, never dies.
Security	Rules shipped in <code>database.rules.json</code> ; every action written to <code>alertLog</code> .
Disruption risk	Additive. The paper register still works if the app does not.
Interoperability	Built on NLEM 2022. Next: HMIS export.

New district, same code. The levers are configuration, not rewrites.

- **NLEM 2022** — the national standard, not a state list. 8 PHCs → 25,000: no architecture change
- **Language is a config pack.** Tamil · Hindi · English shipped; 22 official languages follow the same Speech-to-Text + Gemini path
- **BRICS:** the same forecasting and matching design applies to Brazil's *UBS*, South Africa's *PHC* clinics, and Russia's outpatient network. Drug lists and languages swap; the Risk Score formula does not.

Our reachability sweep is the scaling law: **the further a facility sits from a donor, the more forecasting must carry before transfer can help.** That is the variable a national rollout would tune per state.

THE ASK

Seeking an NHM pilot: 1 district, 50 PHCs, 90 days.

Everything on the previous slides is reproducible from this repository.
Two scripts, one command each, no cloud account required.

[GITHUB.COM/VENKATANARAYANA15/AROGYACHAIN-AI](https://github.com/VENKATANARAYANA15/AROGYACHAIN-AI)

[AROGYACHAIN-AI.WEB.APP](https://aogyachain-ai.web.app)

[DEMO VIDEO](#)

Team: [Your Name] - [Your Role]

“Every PHC that can’t afford to guess, shouldn’t have to.”